Selected Java Kahoot Questions

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1. Which of the following statements about atomic data types is **NOT** true?

- **a) They solve read-modify-write race conditions, but NOT check-then-act race conditions ✓**
- b) They use special CPU instructions (“compare-and-swap”, CAS)
- c) They guarantee thread safety for shared mutable state: single variable
- d) They provide functions implementing certain composite operations atomically

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2. `wait()`, `notify()` and `notifyAll()` are methods of which class/interface?

- a. Thread
- **b) Object ✓**
- c) Runnable
- d) Thread.State

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3. What can happen if two threads try to write to the same resource without synchronization?

- a. Accessing the same resource from multiple threads is not allowed in Java
- b) A race condition will never occur
- **c) A race condition may occur ✓**
- d) A deadlock may occur

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4. Which `Thread` method can make the main thread wait for all other threads to finish?

- a. wait()
- **b) join() ✓**
- c) sleep()
- d) stop()

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5. How many tasks can run in parallel on a machine with 4 processors, each with 2 cores supporting hyper-threading?

- a. 16
- **b) 8 ✓**
- c) 4
- d) Unlimited

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6. Which of the following principles is **NOT** part of the REST architecture?

- a. REST is a client-server architecture
- b) REST offers caching capabilities
- **c) REST is stateful ✓**
- d) REST is a layered system

7. Every resource in REST is defined by?

- a. Socket
- **b) URI (Uniform Resource Identifier) ✓**
- c) A unique ID
- d) None of the above

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8. Which of the following characteristics does **NOT** apply to JSON?

- **a) Difficult to parse ✓**
- b) Concise and intuitive
- c) Readable by both humans and machines
- d) Language-independent specification

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9. Which of the following data types exists in JSON?

- **a) Boolean ✓**
- b) Float
- c) List
- **d) Number ✓**

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10. Which of the following is **NOT** true about the `HttpClient` included in the JDK since Java 11?

- **a) It supports only HTTP/1 ✓**
- b) It supports both synchronous and asynchronous requests
- c) It consists of several classes and interfaces in the `java.net.http` package
- d) All three answers are correct

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11. An `HttpClient` instance in Java is created with `HttpClient client = new HttpClient()`?

- a. True
- **b) False ✓**

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12. Which of the following statements about an `HttpRequest` instance are true?

- a. It is mutable and can be sent multiple times
- **b) It is created via a builder method ✓**
- c) It is immutable and can only be sent once
- **d) The default HTTP method is GET() and can be omitted ✓**

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13. What does the method `sendAsync(HttpRequest, BodyHandler)` of `HttpClient` return?

- a. Callable
- b) Optional
- **c) CompletableFuture ✓**
- d) None of the above

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14. Which of the following array declarations/initializations is **NOT** valid in Java?

- a. `int[] a;`
- b) `int b[];`
- c) `int[] c[] = {{1}};`
- **d) `int d[7];` ✓**

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15. How many distinct String instances will be created after executing the following code?

```
String s1 = "FMI";
String s2 = "FMI";
String s3 = new String("MJT");
```

- a. 1
- **b) 2 ✓**
- c) 3
- d) 4

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16. Which of the following is **NOT** a valid modifier for an interface method in Java?

- a. private
- b) abstract
- **c) final ✓**
- d) static

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17. Is the following statement true: `null instanceof String`?

- a. Yes
- **b) No ✓**

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18. Would we use a `Collection<? super Number>` when we want to **put** elements but not **get** them?

- **a) Yes ✓**
- b) No

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19. How many abstract methods does the `java.io.Serializable` interface contain?

- **a) 0 ✓**
- b) 1
- c) 2
- d) 3

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20. If I/O stream A wraps stream B, does closing A automatically close B?

- **a) Yes ✓**
- b) No

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21. Which of the following statements about `java.util.Optional` are true?

- **a) It is used to avoid null checks or NullPointerException ✓**
- b) It is used to define methods with a variable number of arguments

- c) It represents a container for a value that will be available in the future
- **d) Its instances are returned by some Stream API methods ✓**

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22. How can we make a thread a daemon in Java?

- **a) By calling `setDaemon(true)` ✓**
- b) By calling `setPriority(10)`
- **c) By starting it from another daemon thread ✓**
- d) When the thread that started it finishes execution

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23. Which of the following statements about atomic operations in Java are true?

- **a) They allow avoiding synchronization, making them very fast ✓**
- **b) They are used to prevent race conditions ✓**
- c) They must always be executed in a synchronized section
- d) Two consecutive atomic operations are also an atomic operation

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24. What are the main classes from the `java.net` API for network communication?

- a. SocketChannel
- **b) ServerSocket ✓**
- **c) Socket ✓**
- d) ByteBuffer

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25. Which of the following `java.nio` operations are blocking in non-blocking mode on all channels?

- a. `ServerSocketChannel.accept()`
- b) `SocketChannel.read(ByteBuffer buf)`
- **c) `Selector.select()` ✓**
- d) `Selector.selectNow()`

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26. How is an instance of `java.net.http.HttpResponse` created?

- a. Via a constructor
- b) Via a factory method
- c) Via a builder method
- **d) As a result of calling the `send` method on HttpClient ✓**

26b. Which of the following are valid data types in JSON?

- **a) Boolean ✓**
- b) Function
- c) Undefined
- **d) String ✓**

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27. Which of the following is true about the S.O.L.I.D. **Interface Segregation** principle?

- a. In Java 15 it is achieved via sealed interfaces
- **b) A client should not implement an interface it does not use ✓**

- **c) Several specific interfaces are preferred over one general one ✓**
- d) In Java it is achieved via access modifiers